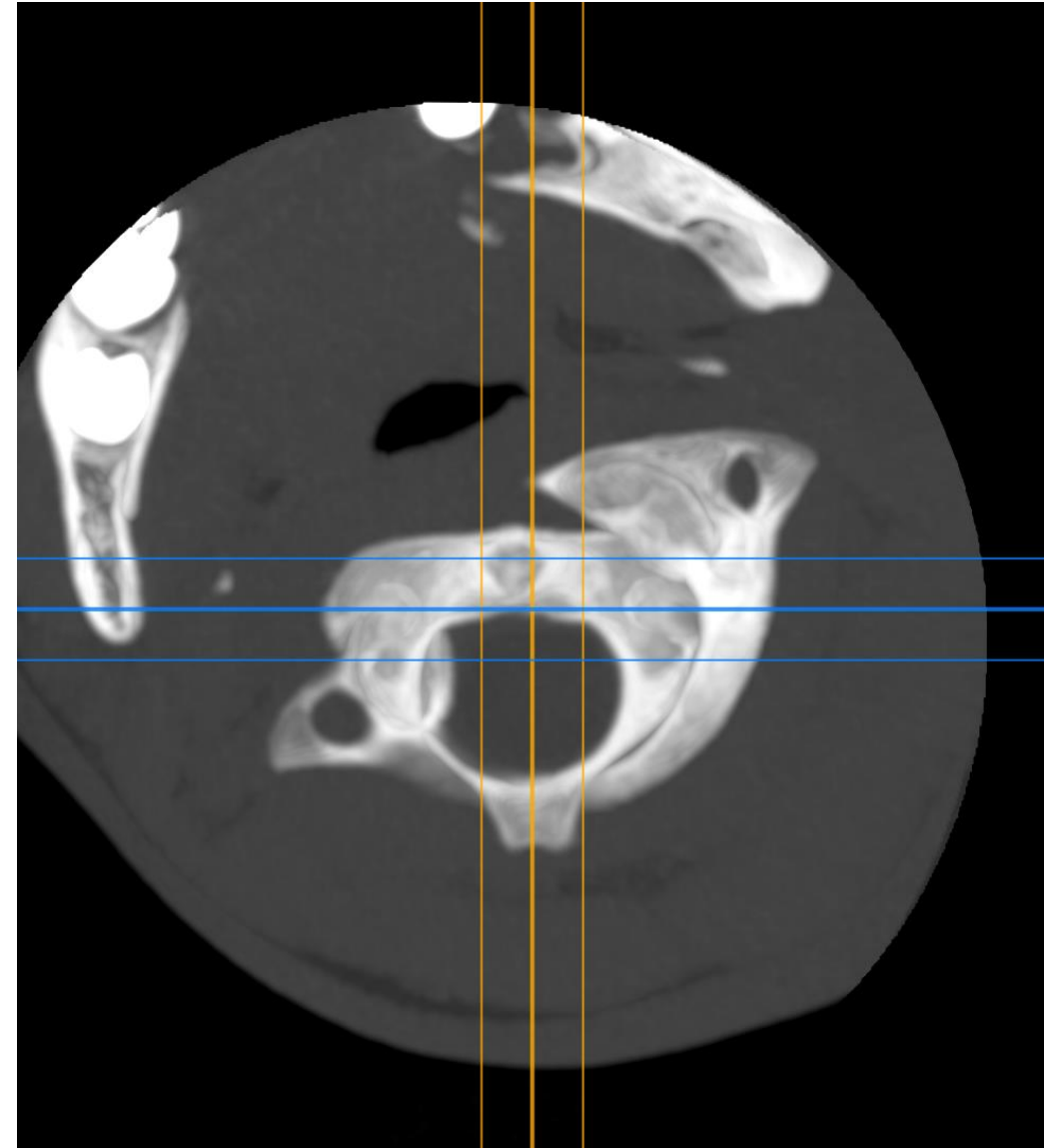


Visual guide to angle measurement

1. Define the head position

- By the side toward which the chin was initially pointing on neutral position
 - Start the measurement with neutral position first
- In this case the chin is pointing right on neural position
 - Right torticollis
- Right is defined as positive from now on

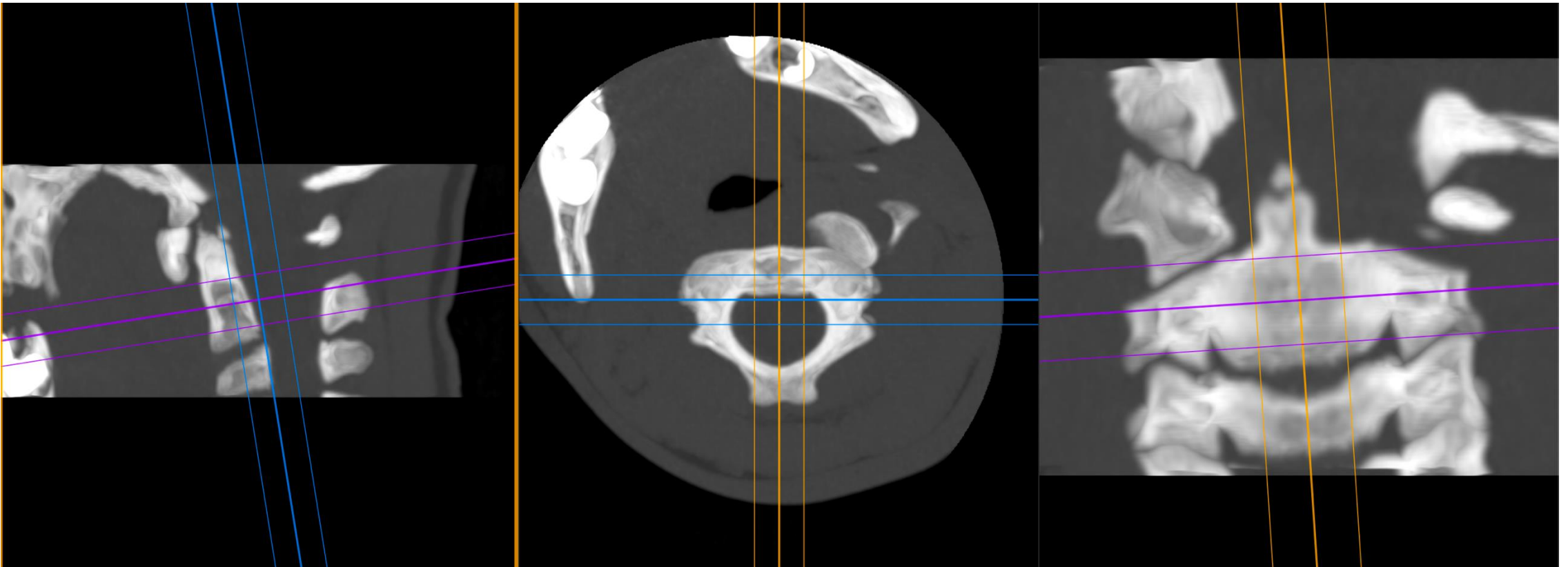


2. MPR reconstruction

- The aim is to consistently measure C0, C1 and C2 angles against a standard reference line to calculate C0-C1 and C1-C2 angles.
- MIP images would be helpful for visualisation of anatomical landmark for measurements
 - (As well for visual assessment whether C1 crosses C2 on different head rotation)

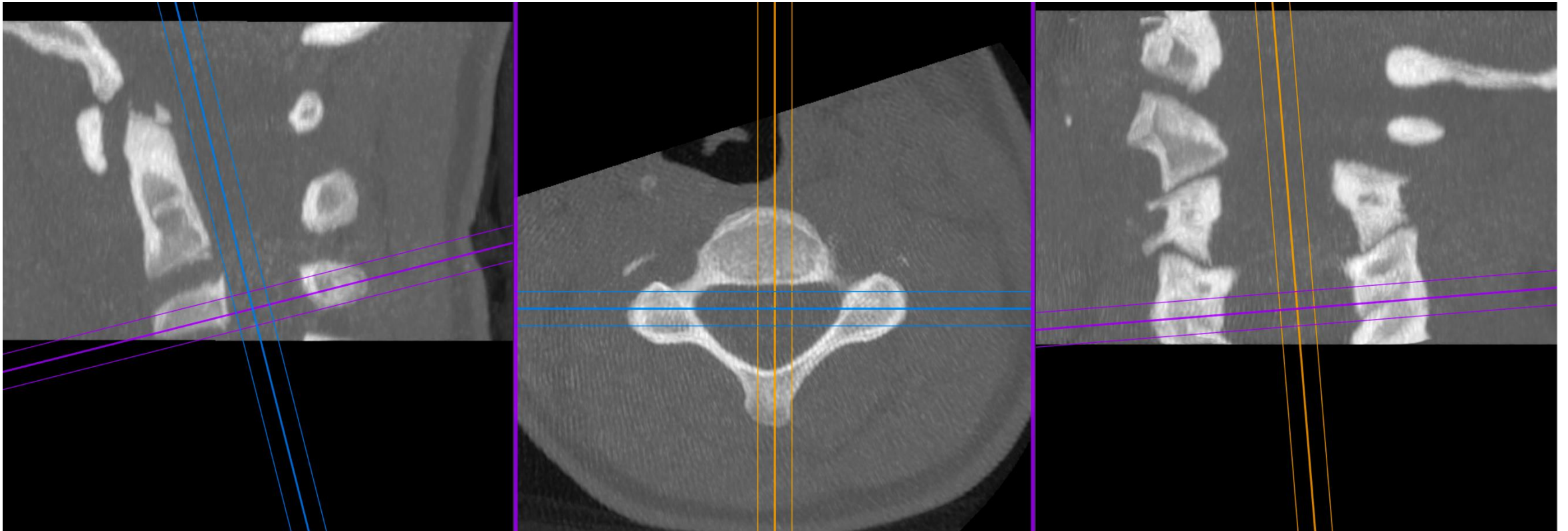
2. MPR reconstruction

1. Fix the measuring plane – preferably in the plane of base of C2 as it is usually consistent



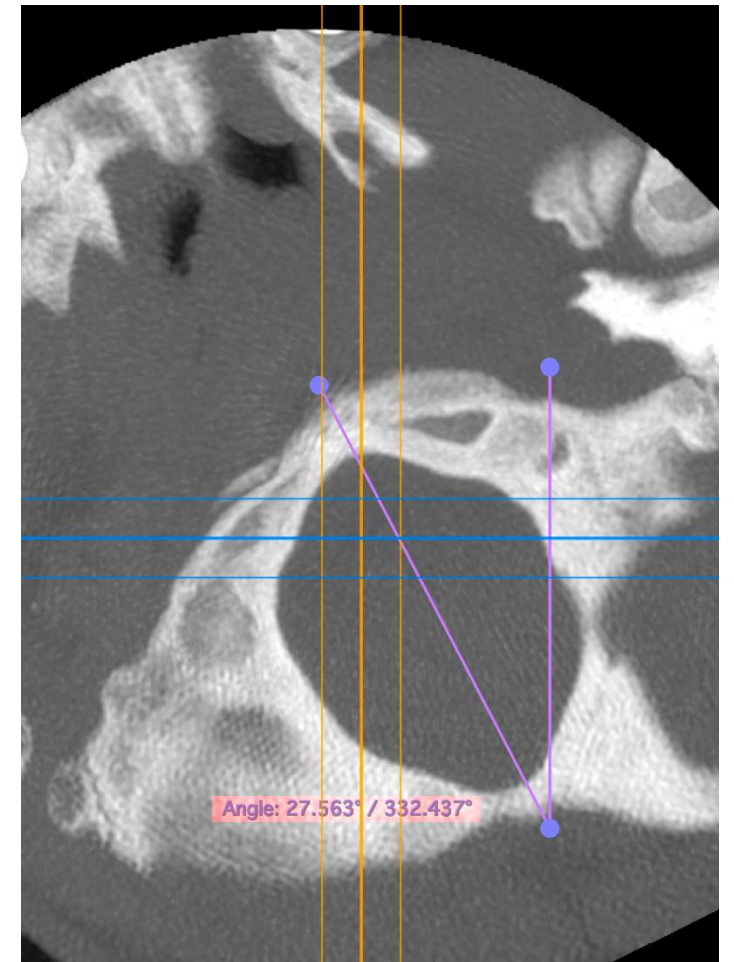
2. MPR reconstruction

2. Rotate the image of the subaxial cervical spine is horizontal.
(For accurate measurement of C1 angle)



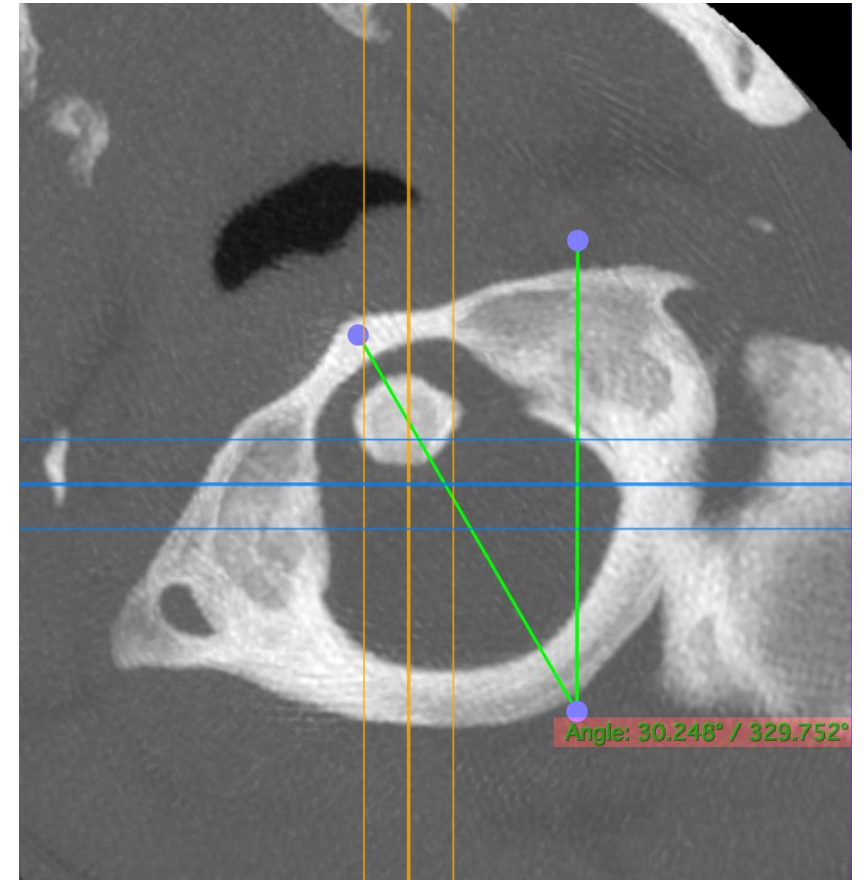
3. Measurement – Neutral position

- Measure C0 angle – Line joining the basion and opisthion of the foramen magnum to a vertical line
- +27.563° in this case (right is positive)



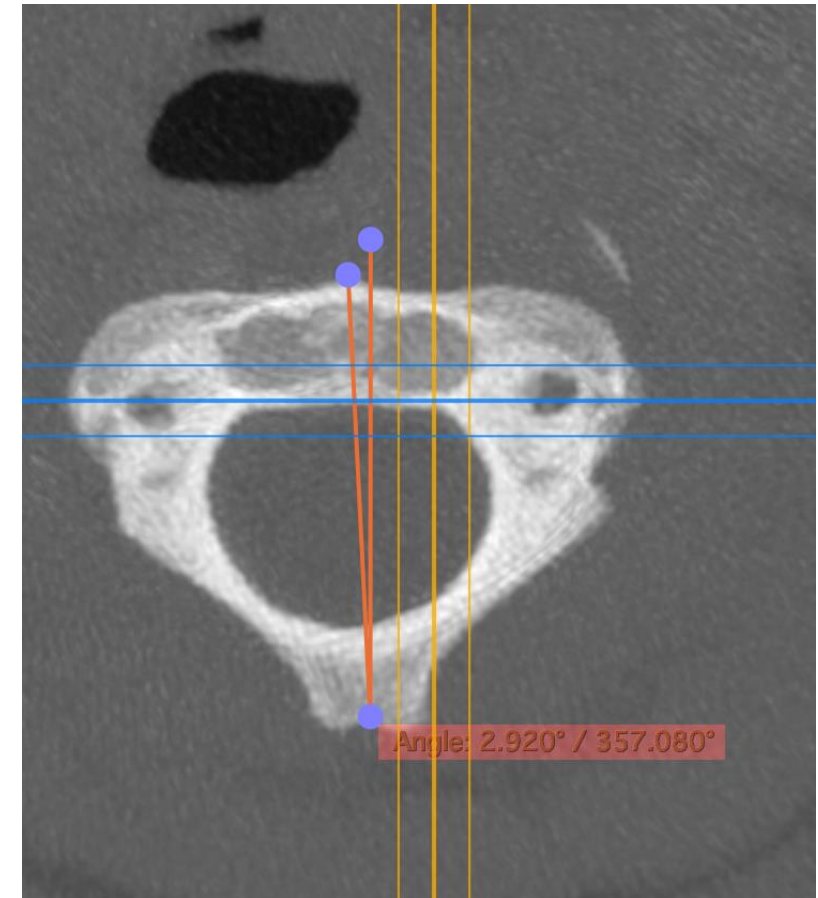
3. Measurement – Neutral position

- Measure C1 angle – Line joining the anterior and posterior tubercle to a vertical line
- +30.248° in this case (right is positive)



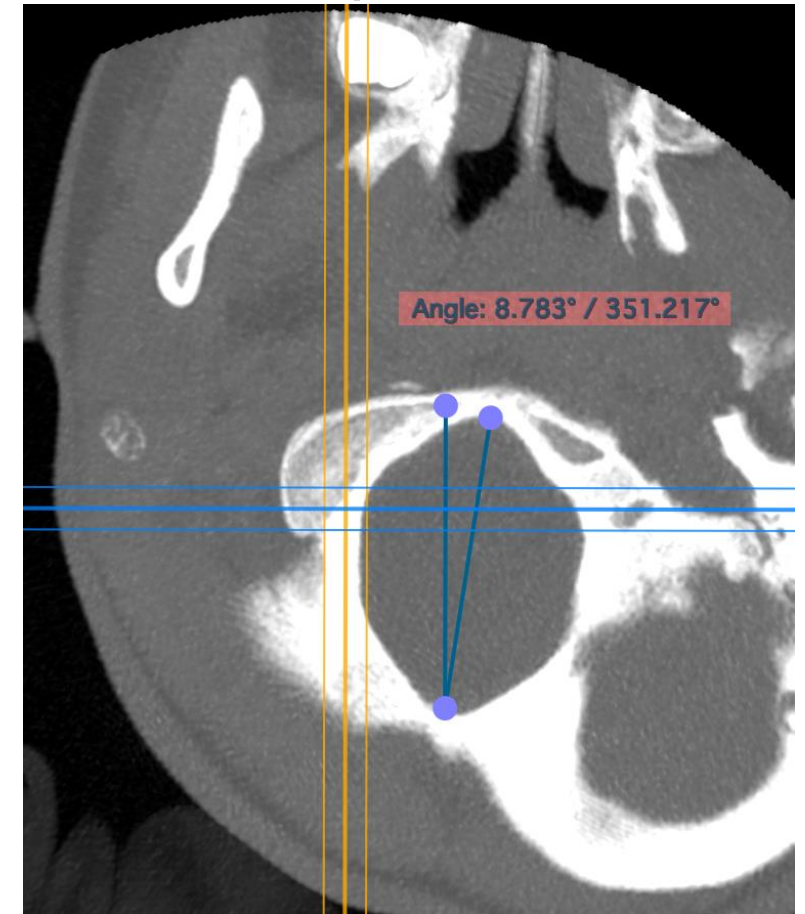
3. Measurement – Neutral position

- Measure the C2 angle - Line joining mid-points of anterior arch and posterior arch to a vertical line
- $+2.920^{\circ}$ in this case (right is positive)



4. Measurements – other positions

- Repeat measurements of C0, C1 and C2 angles in other positions accordingly
- For example, in left position
 - The C0 angle is -8.783° (Right is positive)



5. Tabulation

- Input the measure into the calculator and generate findings.